

IDEA

THEORIES OF RELATIVITY

The principle of relativity dates back to the scientific revolution of the 17th century (see pp.266–67). First identified by Italian physicist Galileo Galilei, it states that physical laws should behave in the same way in all isolated frames of reference unless they are being affected by external forces—in other words, natural laws are the same whether one is in motion or not. This is because all motion is relative—there is no universal, fixed reference point.

Einstein's Special Theory of Relativity (1905) reconciles this principle with the fact that light always travels at the same speed relative to an observer no matter how fast they travel relative to its source. In order to do this, Einstein explained how time and

distance measurements become distorted at high relative speeds. The bizarre results of special relativity are used routinely by experimental physicists.

In 1915, Einstein extended Special Relativity to create his General Theory of Relativity. This theory explains gravity as a distortion of space-time, the four-dimensional “fabric” of the universe.

The two theories of relativity have been tested countless times and have never yet failed. One effect of general relativity (illustrated by this Hubble Space Telescope image) is a phenomenon called gravitational lensing, in which light from distant stars and galaxies is deflected and warped as it passes close to stars and galaxies.

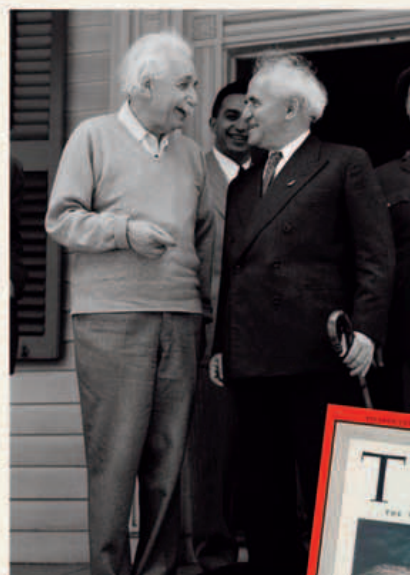
“Do not worry about your difficulties with **mathematics**. I can assure you mine are **still greater**.” ALBERT EINSTEIN

proved that mass and energy, too, are relative quantities, just as time and space are. Furthermore, he showed that mass and energy are two aspects of the same thing, mass-energy. It was in this paper that he derived his most famous equation, relating mass (m) and energy (E), via the square of the speed of light (c^2): $E = mc^2$. This relationship is the basis of nuclear power: inside a reactor, atomic nuclei are broken apart and lose mass, producing large amounts of energy in agreement with Einstein's theory. This famous equation is also the source of the devastating energy of nuclear weapons.

Recognition

In 1907 Hermann Minkowski, one of Einstein's teachers, showed that time is equivalent to the three dimensions of space, and is intertwined with them. He visualized “space-time” as a four-dimensional “fabric” of the universe.

However, few other physicists paid attention to this work at the time. Einstein, meanwhile, set about generalizing his Special Theory of Relativity to include gravity. He realized that time slows down in intense gravitational fields—and one result of this is that light bends as it passes close to very massive objects, such as the Sun. Einstein was able to explain gravity as a



Einstein and politics

Einstein's Zionism brought him in contact with Israel's first prime minister, David Ben-Gurion (above). By 1946, Einstein was already proclaiming the need for nuclear disarmament (right).



distortion of space-time, and published his ideas in 1915, as the General Theory of Relativity.

A total solar eclipse in 1919 provided a chance to test Einstein's new theory. A team of physicists photographed the area of the sky surrounding the Sun during the eclipse. Stars in the dark sky around the obscured Sun appeared slightly out of place—a result of their light bending as it passed close to the Sun. The shift was inexplicable

using the traditional “Newtonian” (see p.269) idea of gravity, but it matched Einstein's theory exactly.

Einstein became a worldwide celebrity. He won the 1921 Nobel Prize for Physics (the first of many honors), and stayed at the forefront of discovery during a period when physicists were beginning to unravel the universe's fundamental mysteries. He was celebrated beyond the academic life of science, becoming a cultural icon, and one of the few scientists known by nearly everyone.

Political life

One consequence of Einstein's fame was that people sought his opinions on a wide range of topics. He was acquainted with many world leaders, wrote widely in newspapers, books, and magazines, and became involved with many political causes. In

his later years, he lobbied against racism in the US and against nuclear weapons. He was a pacifist and vegetarian.

A German-born Jew, Einstein had Zionist (see p.414) aspirations, and was in favor of establishing a Jewish homeland in the Middle East. Having seen the rise of Nazism in Germany at first hand, it was inevitable that he would become involved in the formation of the state of Israel (see pp.414–15). He was even asked if he would become its president.

Although Einstein's theories are only truly understood by physicists, most people recognize that his work radically changed our understanding of the world, and appreciate that this brilliant man was one of the most enduring characters of the 20th century.

TIMELINE

- **March 14, 1879** Albert Einstein is born in the small town of Ulm, Germany. The Einsteins move to Munich the following year.
- **1892** At age 13, Einstein begins his studies at the Lutipold Gymnasium (Munich).
- **1896** In order to avoid military service, he renounces his German citizenship, and begins studying mathematics in Zurich, Switzerland.
- **1901** Einstein graduates from the ETH, a technical institute in Zurich, with a teaching certificate. He is awarded Swiss citizenship.
- **1903** Marries his first wife, Mileva Maric, whom he met at college. He begins work at the patent office in Bern.
- **1905** In Einstein's “Wonderful Year,” he writes four groundbreaking scientific papers, which are published in the German physics journal *Annalen der Physik*. He also gains his doctorate from the University of Zurich.
- **1911** Accepts a professorship at Charles University, Prague. From here, he publishes the first element of his General Theory of Relativity: the idea of gravitational redshift.
- **1912** Returns to Switzerland, becoming a professor at the ETH in Zurich.
- **1914** Einstein becomes director of the Kaiser Wilhelm Institute in Berlin and professor of theoretical physics at the University of Berlin.
- **1915** Publishes his General Theory of Relativity.
- **1917** Einstein publishes an important paper in which he proposes a “cosmological constant,” which would help to explain the expansion of the universe. He also publishes a paper in which he suggests “stimulated emission” could be possible—this is the principle behind the laser, which was not invented until the 1950s.
- **1919** A total eclipse of the Sun on May 29 provides crucial proof for Einstein's General Theory of Relativity. Einstein's divorce is finalized, and he is married for the second time, to his cousin Elsa Löwenthal.
- **1922** Einstein is awarded the 1921 Nobel Prize for Physics “for his services to Theoretical Physics, and especially for his discovery of the law of the photoelectric effect.”
- **1933** Einstein and his wife move to Princeton, New Jersey. Einstein takes up a position at the Princeton Institute for Advanced Study.
- **1939** Writes to US president Franklin D. Roosevelt, warning him that Nazi Germany may be building a nuclear weapon.
- **1940** Becomes a US citizen.
- **April 18, 1955** Dies of heart failure. He is cremated, but beforehand, his brain is removed and preserved.



LECTURING IN THE US